

“For the problem sets, do them. They help and highlight what you need to work on. Before doing them you might be like yeah, I’m good, I know it all. But you don’t and the problem sets help you figure out what you need to work on.” A 2019 Nerd

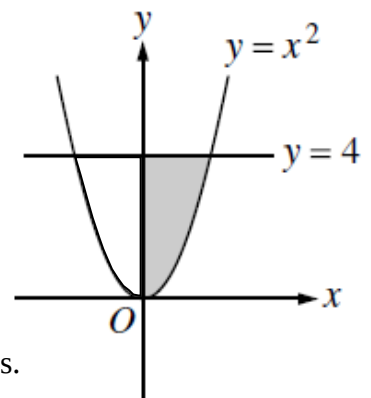
FR1. 2005B BC 5 (No Calculator)

Consider the curve given by $y^2 = 3 + xy$.

- a) Show that $\frac{dy}{dx} = \frac{y}{2y - x}$.
- b) Find all points (x, y) on the curve where the line tangent to the curve has slope $\frac{1}{2}$.
- c) Show that there are no points (x, y) on the curve where the line tangent to the curve is horizontal.
- d) Let x and y be functions of time t that are related by the equation $y^2 = 3 + xy$. At time $t = 5$, the value of y is 3 and $\frac{dy}{dt} = 6$. Find the value of $\frac{dx}{dt}$ at time $t = 5$.

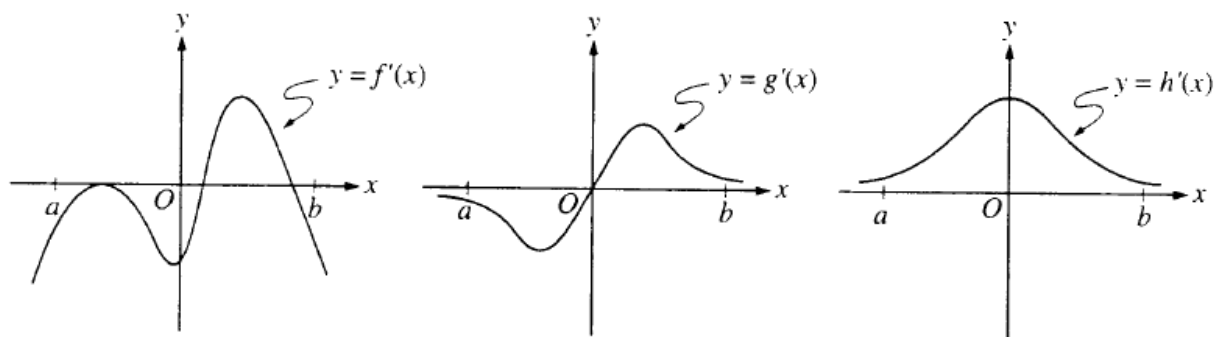
FR2. 1999 BC 2 (No Calculator)

The shaded region R is bounded in the first quadrant by the graph of $y = x^2$ and the line $y = 4$, as shown in the figure to the right.



- a) Find the area of R .
- b) Find the volume of the solid generated by revolving R about the x -axis.
- c) There exists a number k , such that when R is the base of a solid whose cross-sections perpendicular to the x -axis are rectangles of height k , the resulting solid has the same volume as the solid in part b). Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

MC1.



The graphs of the derivatives of the functions f , g , and h are shown above. Which of the functions f , g , or h have a relative maximum on the open interval $a < x < b$?

- (A) f only
- (B) g only
- (C) h only
- (D) f and g only

MC2. If $\int_1^{10} f(x) dx = 4$ and $\int_{10}^3 f(x) dx = 7$, then $\int_1^3 f(x) dx =$

- (A) -3
- (B) 0
- (C) 3
- (D) 10
- (E) 11

MC3. $\lim_{x \rightarrow 0} \frac{7x - \sin x}{x^2 + \sin(3x)} =$

- (A) 6
- (B) 2
- (C) 1
- (D) 0

MC4. The average value of $f(x) = x^2\sqrt{x^3 + 1}$ on the closed interval $[0, 2]$ is

- (A) $\frac{26}{9}$
- (B) $\frac{13}{3}$
- (C) $\frac{26}{3}$
- (D) 13